

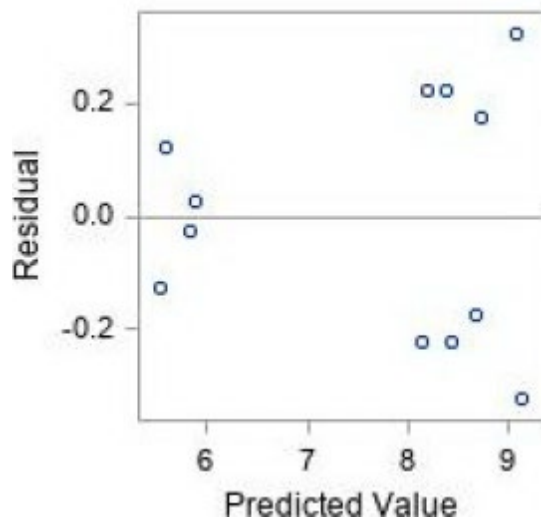
Statistical Analysis of Battery Data

The battery data for the three devices is given in Lecture Notes no 13. The design of experiments used here is a two way factorial with day being a block factor. The model is:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + R_k + \varepsilon_{ijk},$$

where $i=1, 2$ and $j=1, 2, 3$ and $k=1, 2$.

SAS file no 1 for Battery Life data and SAS output no 1 for Battery Life data is our first attempt to analyze this data set. From the Diagnostics Fit plots page is this SAS output we observe that the constant variance of the random errors for the model listed above is not constant as there exists a pattern, increasing triangular one, in the plot of Residual vs. Predicted Value:

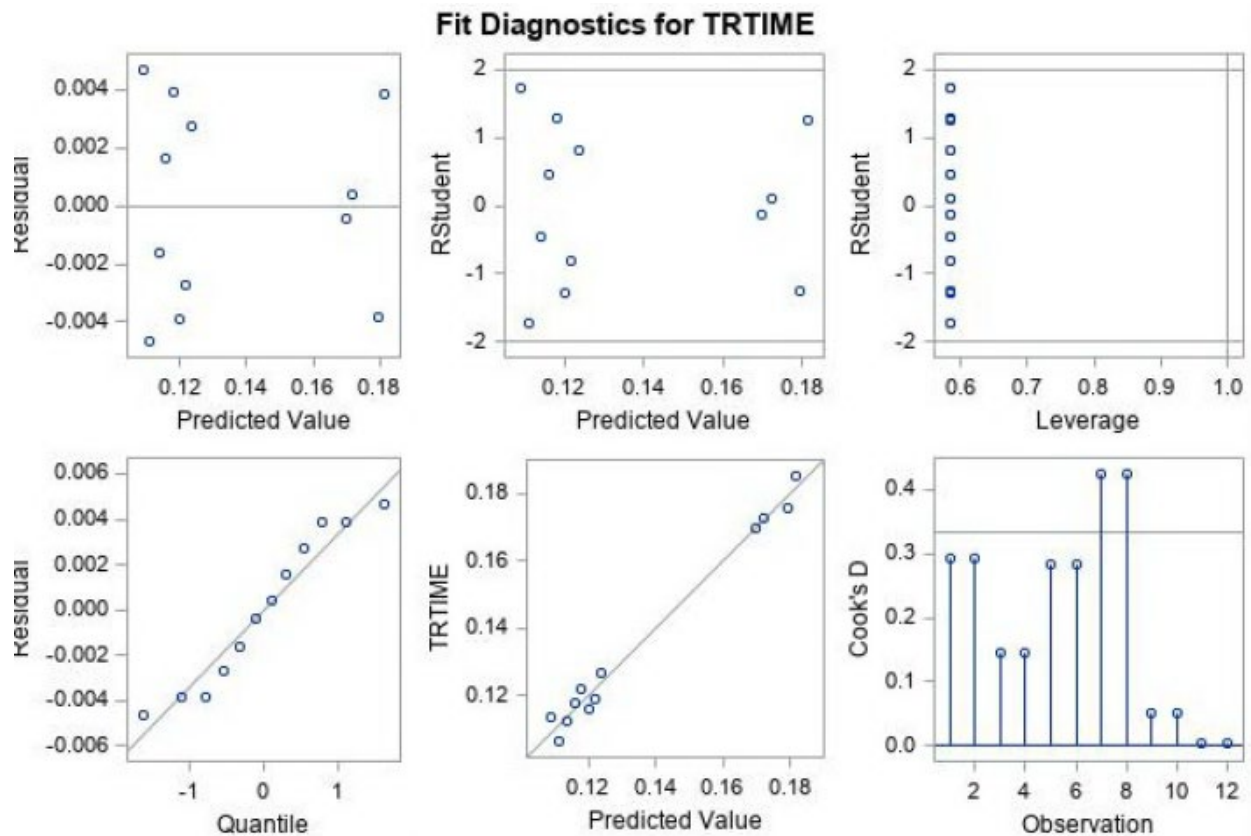


Therefore, the inverse transformation, $1/Y_{ijk}$, is used in the SAS file no 2 for Battery Life data:

$$1/Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + R_k + \varepsilon_{ijk},$$

and the statistical analysis is presented in SAS output no 2.

We first examine the underlying assumptions for the transformed model via the Fit Diagnostics plots and the p-value of the W statistic, presented below:



Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.92448	Pr < W	0.3252

First, we note that there is no pattern in the plot of Residual vs. Predicted Value. The normal probability plot is pretty straight and the p-value of the W statistic is $.3252 > .10$. There are no outliers or influential observations. Therefore, all the underlying assumptions for the transformed model are valid.

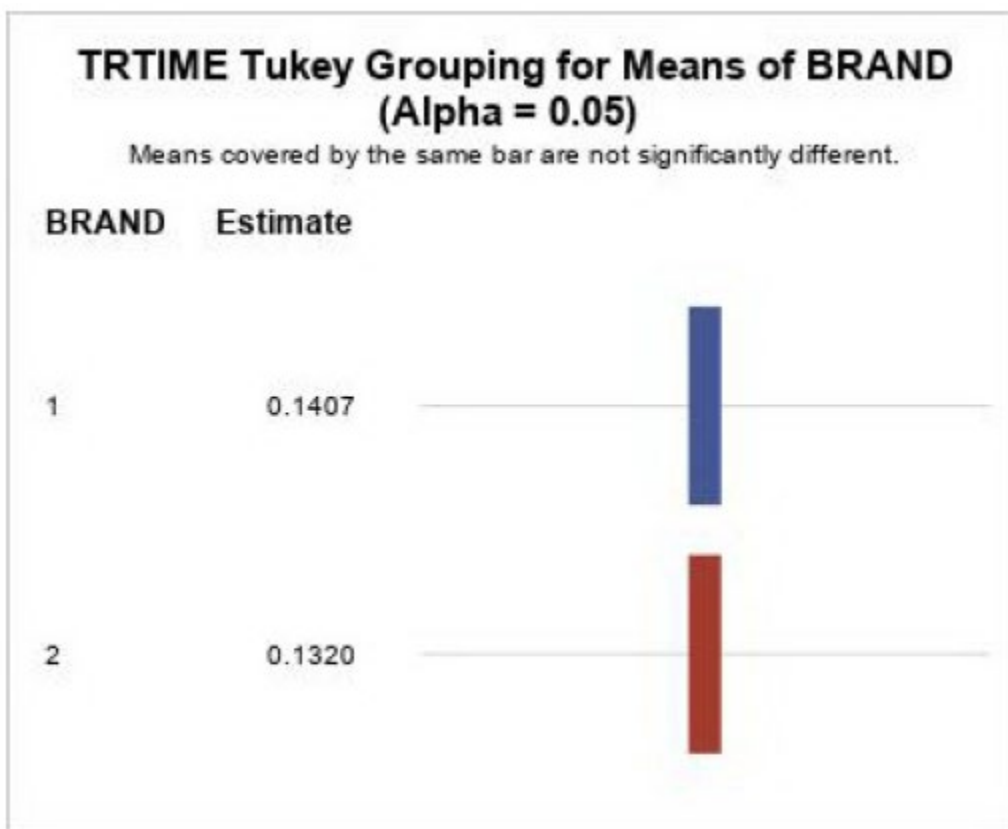
Suppose we would like to perform the tests with an overall significance level of $\alpha = .10$. Since we are interested in testing three null hypotheses, the nominal level is $.10/3 = .033$.

Therefore, based on the Type III SS Table:

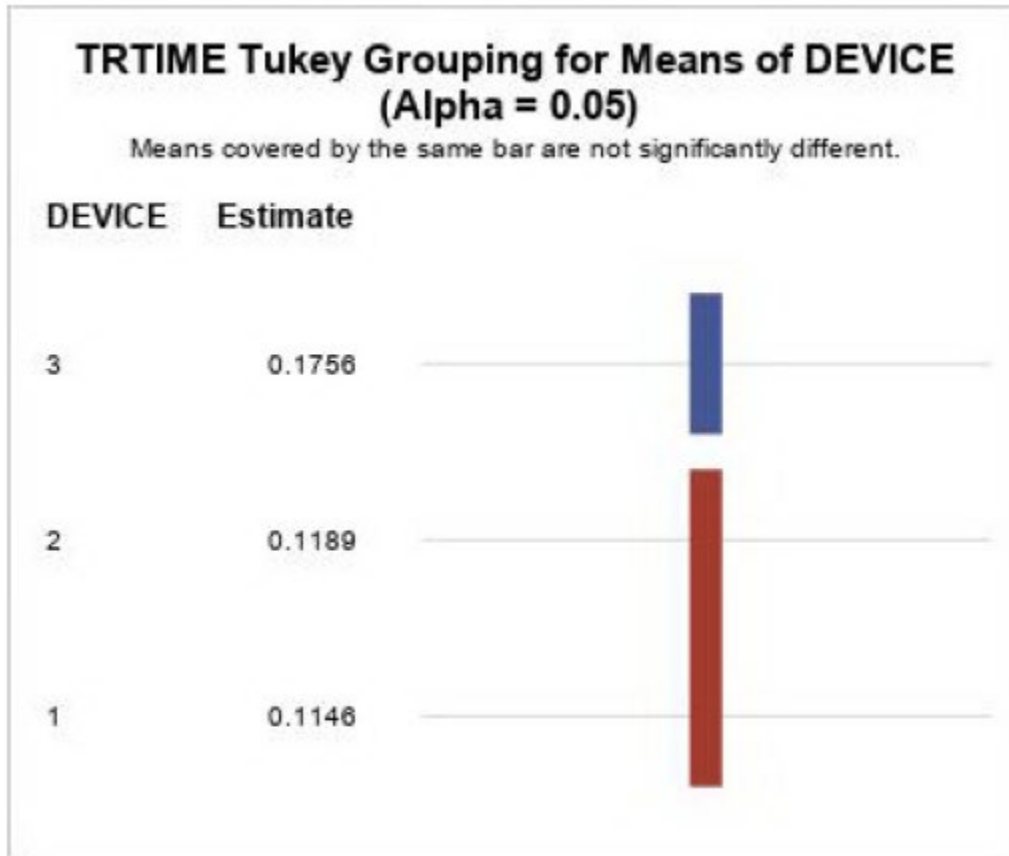
Source	DF	Type III SS	Mean Square	F Value	Pr > F
BRAND	1	0.00023015	0.00023015	9.33	0.0283
DEVICE	2	0.00928792	0.00464396	188.22	<.0001
BRAND*DEVICE	2	0.00000138	0.00000069	0.03	0.9726
DAY	1	0.00001316	0.00001316	0.53	0.4980

we reject the null hypothesis that the two levels the brand of battery have no effect and that the three levels of devices have no effect on the life of battery in the three electronic devices. The interaction between Brand and Device is not significant as far as life of battery is concerned.

Therefore, Tukey multiple comparisons on the two main effects can be performed separately:



and



Since, an inverse transformation has been used brand no 2 of battery has a longer life with an average of $1/.132 = 7.58$ hours. Device 1 (radio) with average life time 8.73 hours and device 2 (camera) with average life time of 8.41 hours have a significant higher operation time than Device 3 (dvd) with average life time of 5.69 hours. The average life times of radio and camera are not significantly different from each other.